



Selected Topics in Cryptography

Lab 04: DSA

March 10, 2026

Develop the following exercises by your own. You can use C/C++, C#, Java or Python to develop your source code, but please use only one of this languages to develop all the exercises. You **MUST NOT USE any cryptographic library to implement DSA.**

1 Key generation for DSA

1. Design and implement a function to generate prime numbers p and q , as follows. Use a function of the programming language of your choice to find a prime number $11 \leq q \leq 1024$. Then try different values of $k > 1$ and compute $p = kq + 1$. Test if p is a prime number (use a function of the programming language to test if an integer is a prime number). If p is a prime number return p and q .
2. Design and implement a function to find a generator g of order q as follows.
 - (a) Compute $e = (p - 1)/q$, i.e. the quotient from dividing $p - 1$ by q .
 - (b) Generate a random number $1 \leq h \leq p - 1$
 - (c) Compute $g = h^e \bmod p$
 - (d) Verify that $2 \leq g \leq p - 1$, if not repeat the process
 - (e) If $g^q \bmod p = 1$, then return g .
3. Design and implement a function to generate a key pair for DSA. Use the functions that you previously did to find the prime numbers p, q and the generator g . Then choose a random number $0 < d < q$ and compute $\beta = g^d \bmod p$. The private key will be d and the public key (p, q, g, β)
4. Generate your own key pairs and upload your public key to the Excel file of public keys (see the link)

2 Signature generation

1. Design and implement a function to generate the signature for an integer $1 \leq m \leq q - 1$. Your function must receive as a parameter the integer m and the public parameters of the public key p, q, g and the private key d . Then do the following

- (a) Choose a secret random key $0 \leq K_E \leq q$.
 - (b) Compute $r = (g^{K_E} \bmod p) \bmod q$
 - (c) Compute $s = (m + dr)K_E^{-1} \bmod q$
 - (d) Print the signature (r, s)
2. Choose an integer $1 \leq m \leq q - 1$ and generate the signature (r, s) for m . Upload the message m and the signature to the Excel file.

3 Verification signature

1. Design and implement a function to verify a signature. Your function must receive as parameters: the public key (p, q, g, β) , the message m and the signature (r, s) . Then do the following:
 - (a) Compute $w = s^{-1} \bmod q$
 - (b) Compute $u_1 = wm \bmod q$
 - (c) Compute $u_2 = wr \bmod q$
 - (d) Compute $v = g^{u_1} \beta^{u_2} \bmod q$
 - (e) If $v == r \bmod q$ the signature is valid, otherwise it is not.
2. Use the data listed in the Excel file: m and (r, s) to test your function to verify the signature.

4 Questions

Answer the following questions

1. Use the mathematical expressions of DSA involved in key generation and signature generation to show that if the signature is valid, then $v == r \bmod q$. Write down all the details.
2. Suppose that you are an adversary. You have several pairs of messages m and the corresponding signatures (r, s) . Suppose the same K_E was used to sign these messages, using the same private key d . How can you find d ? Use the mathematical expressions involved in key generation and signature generation and give an algorithm to find d .

5 Products

Every student can write her/his own report including the following information:

1. Personal information, date of the lab session and the topic that we are studying in this lab session.
2. Source code to generate a key pair for DSA
3. Your results to verify the signatures in the Excel file, i.e a list including m , (r, s) and if the signature is valid or not.
4. Answers to the questions of Section 4: Questions.